MIL-C-45192B
17 June 1983
SUPERSEDING
MIL-C-45192A
4 May 1964

# MILITARY SPECIFICATION

## CONSTRICTOR, BEAD EXPANDING, PNEUMATIC TIRE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of manually actuated pneumatic tire constrictor used to expand the beads of deflated tubeless tires against the tire rim to form an airtight seal.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| PPP-B-601 | - | Box, Wood, Cleated Plywood |
| PPP-T-76 | - | Tape, Packaging, Paper (For Carton Sealing) |

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of |
| MIL-B-121 | - | Barrier Material, Greaseproofed, Waterproofed, Flexible |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Sheet, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete tire constrictor for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The tire constrictor shall be new and of standard commercial design. The constrictor shall be a self-contained, manually operated unit composed of a force-exerting mechanism with a contraction band or bands. The design of the constrictor shall permit operation by one person.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the tire equipment industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissable to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The constrictor shall be complete and so constructed that it can be used for all operations for which it was designed.

3.4.1 Castings. Castings shall be free from blow-holes, porosity, hard spots, shrinkage defects, cracks, or other defects. Strength and other physical and chemical properties of the castings shall be adequate to meet the performance requirements specified herein.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.5 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without the need for modification of the parts or the constrictor.



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3.4.6 Surfaces. All surfaces of the tire constrictor shall be clean and free of any harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.7 Safety. Safety devices shall be provided to eliminate safety hazards. Except as otherwise specified (see 6.2), the constrictor shall comply with applicable standards of OSHA 2206.

3.5 Performance.

3.5.1 Functional requirements. The tire constrictor shall transmit a uniform pressure around the tread crown circumference, forcing the beads against the wheel rim to form an air seal and permit inflation of the tire. The unit shall not cause damage to the tire during the bead expanding operation. A controlled release shall be provided to permit safe removal from the expanded bead position.

3.5.2 Capacity. The tire constrictor shall have a capacity that shall provide for the bead expansion of all tubeless tire sizes ranging from 6.50 inches by 13 inches through 9.00 inches by 22.50 inches inclusive (24.50 inch through 38.25 inch diameters) and such other large sizes (off-the-road tires) as specified (see 6.2).

3.5.3 Shock. The tire constrictor and all components shall withstand, without damage, shock resulting from being dropped vertically five times a distance of 36 inches to a concrete floor.

3.5.4 Imposed load and unit removal. The tire constrictor shall be capable of withstanding, without damage, an imposed load of 15 pounds per square inch gage (psig) for a period of ten-minutes. Following the ten-minute load period, the constrictor shall be safely released and removed while the tire is still inflated.

3.6 Components.

3.6.1 Pressure-exerting device. The pressure-exerting device shall be a hand lever or hand-crank actuated mechanical device for contracting the band(s) about the tire tread circumference. The pressure-exerting mechanism shall incorporate a mechanical advantage to permit the bead expanding operation to be performed with a force of not more than 40 pounds applied to the hand lever or hand crank. The pressure-exerting device shall securely retain the band(s) in the contracted position without slippage until released. The release of the pressure-exerting force shall be accomplished without endangering the safety of the operator or damaging the constrictor, tire, or wheel.

3.6.2 Contraction band(s). The contraction band or bands shall be made of steel or nylon webbing and shall be capable of being adjusted and centered about the tread crown circumference of the tire sizes specified. When contracted by the pressure-exerting device, the band(s) shall impose a uniform compression on the crown to expand the beads evenly. The band(s) shall withstand the imposed loads without distortion or breakage. The tire-engaging surface of the contraction band(s) shall be smooth.



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3.7 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Product identification. The tire constrictor shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the tire constrictor shall have the stability, strength, safety, and efficient operating characteristics found in commercial units and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.



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shall be 5. The minimum angular separation between drill particles shall

| CHARACTERISTICS | REQUIREMENT PARAGRAPH |
|---|---|
| Design not as specified | 3.2 |
| Material not as specified | 3.3 |
| Construction defects | |
| Castings | 3.4.1 |
| Welding, brazing, and soldering | 3.4.2 |
| Fastening devices | 3.4.3 |
| Threads | 3.4.4 |
| Interchangeability | 3.4.5 |
| Surfaces and edges | 3.4.6 |
| Components missing or not as specified | |
| Pressure-exerting device | 3.6.1 |
| Contraction band(s) | 3.6.2 |
| Painting not as specified | 3.7 |
| Product identification not as specified | 3.8 |
| Workmanship defects | 3.9 |

4.6 Tests.

4.6.1 Performance test. In accordance with the manufacturer's handbook of instructions, the tire constrictor shall be contracted to expand the beads of four new tires, selected from the largest and smallest capacities specified in 3.5.2 or in the contract. With the beads in the expanded position, the tire shall be inflated until the beads and wheel rim form and maintain the air seal. The constrictor shall then be released and removed. During and upon completion of the test operations, the tire constrictor and the tires shall be observed and examined to determine compliance with the following:

(a) One-person operation (see 3.2).
(b) Unit function (see 3.5.1).
(c) Tire capacity (see 3.5.2).
(d) Operation and mechanical advantage of pressure-exerting device (see 3.6.1).
(e) Contraction band operation (see 3.6.2).

4.6.2 Shock test. The tire constrictor and all components shall be dropped five times to a concrete floor from a height of 36 inches. On conclusion of the test, the tire constrictor shall be examined for evidence of damage and tested in accordance with 4.6.1 to determine compliance with 3.5.3.

4.6.3 Pressure load test. In accordance with the manufacturer's handbook of instructions, the tire constrictor shall be contracted to the bead expanded position on any tire size within the range specified in 3.5.2 or in the contract. The tire shall then be inflated to an air pressure of 15 psig and retained under this imposed load for a period of ten minutes. Upon conclusion of this period, the constrictor shall be released and removed while the tire is still inflated. The tire constrictor shall be examined during and upon conclusion of the test operation to determine compliance with 3.5.4.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.



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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The tire constrictor shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 of MIL-P-116.

5.1.1.2 Preservation. Preservatives used and procedure of application shall be in accordance with MIL-P-116. Immediately after cleaning and drying, all unpainted or unplated ferrous metal surfaces of the item shall receive an application of type P-11 preservative.

5.1.1.3 Unit packs. The unit protection shall be method I of MIL-P-116. Publications issued with the unit shall be packaged method IC-1 of the same specification. The preserved area of the item shall be wrapped with barrier material conforming to MIL-B-121, type 1, grade A, class 2. Secure the wrap with tape conforming to PPP-T-76.

5.1.2 Industrial. Industrial preservation and packaging shall be as specified in ASTM D3951-82.

5.2 Packing. Packing shall be Level A or Industrial, as specified (see 6.2).

5.2.1 Level A. The individual items shall be packed in shipping containers conforming to PPP-B-601, overseas type. Blocking, bracing, anchoring, and cushioning, as applicable, shall be in accordance with MIL-STD-1186. Closure shall be as specified in the container specification.

5.2.2 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.3 Marking. Marking shall be Level A or Industrial as specified (see 6.2).

5.3.1 Level A marking. Level A marking shall be as specified in MIL-STD-129.

5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The constrictor covered by this specification is intended for use in field and base shops. The constrictor expands the beads of tubeless tires against the tire rim to form an airtight seal to permit tire inflation.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).



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(c) Safety requirements if different (see 3.4.7 and 6.3).
(d) Special capacities (large tire sizes) required (see 3.5.2).
(e) Painting and finishing if different (see 3.7).
(f) Product marking if different (see 3.8).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling when required (see 4.4.1).
(j) Level of preservation, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Air Force - 99

Review activities:

Army - AT SM

Preparing activity:

Army - AL

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